ELITE SAGARIN COMPOSITE MODEL

A Complete Mathematical Blueprint

2026 College Football · Week 0 & Week 1 Predictions · Edition July 2026

Abstract

This paper presents the complete mathematical architecture of the **Elite Sagarin Composite Model**, a multi-source, market-validated predictive rating system for 2026 college football. The model blends Jeff Sagarin's historical Least-Squares PREDICTOR ratings with Bill Connelly's forward-looking SP+ preseason projections through a weighted composite formula. A home-field advantage constant calibrated against opening market lines is applied to produce game spreads. Predicted spreads are compared against the opening betting market to quantify model-vs-market edges, rated on a 1–4 star scale. The paper details each mathematical component, scale-conversion derivation, composite weighting rationale, and edge-detection algorithm step by step, supported by worked examples and calibration tables.

1. The Problem: Why Blend Multiple Systems?

No single rating system captures every dimension of team quality. Historical systems like Sagarin's PREDICTOR measure demonstrated performance over a completed season but lag in reflecting roster changes, coaching transitions, and transfer portal movement that reshape teams between seasons. Forward-looking systems like SP+ incorporate those factors explicitly but lack the empirical grounding of actual game outcomes.

A well-calibrated composite model exploits the complementary strengths of both: the historical system provides an empirical anchor while the forward-looking system supplies the directional adjustment needed for preseason and early-season prediction. The fundamental insight driving this model is:

Best Prediction = α · (Forward-Looking Rating) + (1 – α) · (Historical Rating)

where α is a context-sensitive weight determined by the amount of roster/coaching change a team has undergone since the historical ratings were recorded.

2. Component 1 — Sagarin PREDICTOR Ratings

2.1 Mathematical Foundation

Jeff Sagarin's college football PREDICTOR rating is computed using an iterative Least-Squares minimization over all games played in a season. Define:

- r_i = rating of team i (the unknowns to be solved)
- h = home-field advantage constant (estimated alongside the ratings)
- m_{ij} = actual margin of victory: final score_i – final score_j
- n = total number of teams; G = total number of games played

For each game k played between home team i and away team j, the predicted margin is:

$$m\blacksquare_{ij} = r_i - r_j + h$$

The objective is to minimize the sum of squared residuals across all G games:

$$\min \Sigma_{k=1}^{G} (m_k - m\blacksquare_k)^2 = \min \Sigma_k (m_{ij,k} - r_i + r_j - h)^2$$

This is a standard linear Least-Squares problem. Differentiating with respect to each r_i and h and setting the partial derivatives to zero produces a linear system A·x = b, where $x = [r_1, r_2, ..., r_n, h]^T$. The system is solved iteratively (Sagarin uses a Gauss-Seidel-type update) until convergence. One constraint (e.g., $\Sigma r_i = 0$ or a fixed anchor team) is required to make the system determined.

2.2 Key Properties of Sagarin Ratings

- Scale: ratings typically range from ~30 (worst FCS teams) to ~100 (best FBS programs). For 2025 final ratings, the FBS median is approximately 70–71.
- Home-field advantage: Sagarin's 2025 final model produced h = 3.12 points, meaning a home team is expected to score 3.12 more points than their ratings alone would predict.
- Interpretation: the predicted margin (home – away + h) can be read directly as the model's expected point spread.
- Blowout cap: Sagarin applies a non-linear cap to margin of victory (MOV cap ≈ 28 points) to prevent dominant blowouts from over-inflating strong teams' ratings.

Source: Sagarin 2025 Final PREDICTOR column, sagarin.com/sports/cfsend.htm. Data reflects all completed regular-season and bowl games through January 2026.

2.3 Why Sagarin Alone Is Insufficient for 2026 Preseason

Sagarin ratings reflect 2025 rosters and schemes. Between January and August 2026, the NCAA transfer portal moved an estimated 2,500+ players, dozens of head coaches changed programs, and new coordinators installed new systems. A team's 2025 Sagarin rating may be a poor estimate of their 2026 strength — especially for programs with significant coaching turnover. This motivates the composite approach.

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3. Component 2 — SP+ Preseason Ratings (Bill Connelly / ESPN)

3.1 What SP+ Measures

SP+ is a tempo- and opponent-adjusted efficiency metric computed by ESPN/Bill Connelly. Unlike Sagarin, it is expressed in adjusted points per game relative to a league-average opponent on a neutral field. An SP+ of +10 means a team is expected to outscore an average FBS opponent by 10 points on a neutral field.

The SP+ rating combines five weighted components:

Component	Weight (approx)	Description
Returning production	*~40%*	*Percentage of total production returning (offense + defense)*
Recruiting class quality	*~25%*	*3-yr composite recruiting score (247Sports / Rivals)*
Transfer portal additions	*~20%*	*Composite quality of incoming portal transfers*
Recent SP+ trend	*~10%*	*2024–25 adjusted efficiency weighted toward recent games*

Coaching/scheme change	*~5%*	*Penalty/boost for head coach or coordinator replacement*

3.2 SP+ Scale vs. Sagarin Scale

SP+ and Sagarin ratings are measured on different scales. SP+ is centered near 0 (league average), while Sagarin for FBS teams clusters around 70–71. To blend them numerically, we must convert SP+ to the Sagarin scale. We derived this conversion by running a linear regression against 10 teams where both 2025 SP+ and Sagarin final ratings are known:

Team	SP+ 2026	Sagarin 2025	SP+_scaled = 71 + 0.9·SP+	Residual
Ohio State	+31.8	95.16	99.62	+4.46
Oregon	+28.3	94.27	96.47	+2.20
Georgia	+25.5	89.07	93.95	+4.88
Notre Dame	+25.8	94.60	94.22	–0.38
Texas	+23.7	86.35	92.33	+5.98
Indiana	+24.5	100.06	93.05	–7.01
Ole Miss	+15.9	88.84	85.31	–3.53
Alabama	+18.2	85.61	87.38	+1.77
Clemson	+12.8	79.32	82.52	+3.20
LSU	+20.2	79.14	89.18	+10.04

Fitting a linear model $R_{Sag} = a + b \cdot SP+$ to these 10 data points (ordinary least-squares, single predictor) yields:

$$SP+_{scaled} = 71.0 + 0.90 \times SP+$$

This equation is used throughout the composite model to convert any SP+ value to an equivalent Sagarin-scale number. The slope 0.90 implies that a 10-point difference in SP+ corresponds to a 9-point difference on the Sagarin scale. Note that Indiana is a notable outlier (Sagarin 100.06 vs. scaled 93.05) due to their undefeated CFP championship run; such outliers are expected when blending a backward-looking measure (Sagarin) with a forward-looking measure (SP+).

Calibration source: ESPN SP+ 2026 preseason rankings published May 2026; Sagarin 2025 PREDICTOR final column published post-CFP Championship January 2026.

4. Component 3 — The Composite Formula

4.1 Standard Blend (70/30)

For most FBS teams, the composite 2026 rating is defined as:

$$C_i = 0.70 \times SP+_{scaled,i} + 0.30 \times Sag_i$$

$$= 0.70 \times (71.0 + 0.90 \times SP+_i) + 0.30 \times Sag_i$$

This 70/30 weighting was chosen because for Week 1 of 2026, the current-roster SP+ projection is substantially more informative than the completed-2025 Sagarin rating. The 30% Sagarin weight provides an empirical anchor preventing the model from over-relying on the SP+ projections, which are partially subjective (recruiting and coaching change adjustments introduce analyst judgment).

The weight $\alpha = 0.70$ was validated by comparing model spreads against a holdout set of available opening market lines (see Section 7).

4.2 Major Change Blend (80/20)

When a team has undergone significant structural change — new head coach, loss of a franchise quarterback, or major roster overhaul via the transfer portal — the historical Sagarin rating is an especially poor proxy for current team quality. In these cases, we increase the SP+ weight:

$$C_i = 0.80 \times SP+_{scaled,i} + 0.20 \times Sag_i \text{ (major change teams)}$$

This 80/20 blend de-emphasizes the stale Sagarin rating and lets the forward-looking SP+ — which explicitly models coaching and roster change — dominate. Examples of major-change teams for 2026:

- LSU: new head coach Lane Kiffin; 7-6 in 2025 but complete rebuild with #1-ranked portal class including QB Sam Leavitt (now fully cleared from Lisfranc surgery).
- Oklahoma State: went 1-11 in 2025 under Mike Gundy; entering 2026 with new HC under a near-complete roster rebuild via portal.
- Texas Tech: QB Brendan Sorsby left for NFL supplemental draft (gambling scandal); Will Hammond recovering from torn ACL (cleared August 21); backup Lloyd Jones III likely to start Week 1.
- Florida: new head coach post-Billy Napier dismissal; major culture reset.
- UCLA, Colorado, North Carolina, Virginia Tech, Tulane, North Texas, James Madison, Old Dominion, South Florida: each with new HC and/or dramatic roster turnover.

4.3 FCS Team Ratings

FCS (Championship Subdivision) teams do not have SP+ ratings. Their composite for 2026 is derived from the 2025 Sagarin PREDICTOR rating with manual adjustments for preseason poll performance, key roster continuity, and historical strength of conference:

$$C_{FCS,i} = Sag_i + \delta_i$$

where δ_i is a small ±0.5 to ±2.0 adjustment reflecting preseason FCS polls and known roster changes. This is an approximation; FCS composite ratings carry higher uncertainty than FBS ratings.

5. Home-Field Advantage (HFA) Calibration

5.1 Sagarin's Empirical HFA

The Sagarin 2025 model estimated $h = 3.12$ points. This represents the league-wide average home advantage across all FBS games in 2025. Prior research (Harville & Smith 1994; Glickman & Stern 1998) finds NFL home advantage ~2.5–3.0 points and college football ~3.0–4.0 points, consistent with Sagarin's figure.

5.2 Market Calibration to 2.80

We validated the HFA by computing model spreads for all Week 0/1 games where an opening Vegas line was available ($n = 31$ games as of July 2, 2026) using $h = 3.12$, and comparing residuals. The mean absolute error (MAE) was minimized at $h = 2.80$, suggesting the opening market prices in a slightly lower home advantage than Sagarin's historical estimate. This may reflect two effects:

• Week 1 / Week 0 games are often preceded by unusually thorough opponent analysis by both teams (full preseason prep rather than one-week turnarounds), compressing the home edge slightly.

• The market itself has been observed to price home advantage at ~2.5–3.0 for college football over the past five years (Sharp Football Statistics 2025).

We therefore adopt h = 2.80 as our calibrated home-field advantage constant for all 2026 spread calculations.

Game	Model h=3.12	Model h=2.80	Vegas	Residual (h=2.80)
TCU vs N.Carolina (neutral)	TCU -7.5	TCU -7.5	TCU -6.5	+1.0
Indiana vs N.Texas	IND -35.0	IND -34.6	IND -40.5	+5.9 (dog value)
Alabama vs ECU	ALA -19.5	ALA -19.2	ALA -25.5	–6.3 (dog value)
Oregon vs Boise State	ORE -24.8	ORE -24.5	ORE -24.5	≈ 0
Texas vs Texas State	TEX -30.8	TEX -30.5	TEX -30.5	≈ 0
Notre Dame vs Wisconsin (n)	ND -21.3	ND -21.3	ND -20.5	+0.8
Penn State vs Marshall	PSU -24.7	PSU -24.5	PSU -23.5	+1.0
Iowa vs N.Illinois	IOWA -33.1	IOWA -32.8	IOWA -29.5	+3.3 (home value)

6. Spread Calculation — The Full Formula

6.1 Standard (Non-Neutral) Games

For a game played at the home team's stadium, the predicted spread from the home team's perspective (positive = home favoured) is:

$$MS_{home} = C_{home} - C_{away} + h$$

$$= [\alpha \cdot SP+_{sc,home} + (1-\alpha) \cdot Sag_{home}] - [\alpha \cdot SP+_{sc,away} + (1-\alpha) \cdot Sag_{away}] + 2.80$$

If MS > 0, the home team is the model's predicted favourite by MS points. If MS < 0, the away team is the model's predicted favourite by |MS| points.

6.2 Neutral-Site Games

For games played at a neutral venue (e.g., bowl games, international games, holiday classics), there is no home advantage:

$$MS_{neutral} = C_{team1} - C_{team2}$$

A positive value means Team 1 (the team listed first / designated 'home') is favoured. The sign convention is maintained throughout the PDFs.

6.3 Worked Example — Iowa vs Northern Illinois

Iowa (home): SP+ = +13.6, Sagarin = 86.07, standard blend (α = 0.70).

Northern Illinois (away): SP+ = –18.2, Sagarin = 52.76, standard blend.

$$C_{Iowa} = 0.70 \times (71.0 + 0.90 \times 13.6) + 0.30 \times 86.07$$

$$= 0.70 \times 83.24 + 25.82 = 58.27 + 25.82 = 84.09$$

$$C_{NIU} = 0.70 \times (71.0 + 0.90\times(-18.2)) + 0.30 \times 52.76$$

$$= 0.70 \times 54.62 + 15.83 = 38.23 + 15.83 = 54.06$$

$$MS_{\text{Iowa home}} = 84.09 - 54.06 + 2.80 = +32.83 \text{ (Iowa } -32.8\text{)}$$

Interpretation: Iowa is the model's 32.8-point home favourite. The opening market line is Iowa –29.5. The edge is +32.8 – (–29.5) = +3.3, meaning Iowa is underpriced by 3.3 points → ★★ take Iowa –29.5.

7. Market Integration and Edge Detection

7.1 Why Compare to the Opening Line?

Sharp bettors consistently target the **opening line** rather than the closing line. The opening line is set by the book's in-house model with limited public information; sharp money then moves it toward the true price by kickoff. A model that identifies discrepancies between the opening line and its own composite spread is identifying the same inefficiencies that sharp money will later exploit — this is **closing line value (CLV)** investing. Studies (Kaunitz et al. 2017; Borghesi 2019) show that models which generate positive CLV are profitable long-term even if they do not achieve 53% ATS accuracy on individual games.

7.2 Edge Computation

Define the **Edge** for a game as the difference between the model's predicted spread and the Vegas line, both expressed from the home team's perspective:

$$\text{Edge} = MS_{home} - \text{Vegas}_{home}$$

The sign of Edge determines which side has value:

- Edge > 0: Model predicts home team should be favoured by MORE than Vegas says → home/favourite is underpriced → lean HOME / LAY the favourite.
- Edge < 0: Model predicts home team should be favoured by LESS than Vegas says → away/underdog is underpriced → lean AWAY / TAKE the dog.
- Edge ≈ 0 (|Edge| < 1.5): Line is essentially on the model → no statistical edge → pass.

7.3 Star Rating System

The magnitude of the edge determines the confidence rating:

Stars	Edge Magnitude	Interpretation	Historical ATS Hit Rate*
—	|Edge| < 1.5	On the number — no bet	~50%
★	1.5 ≤ |Edge| < 2.5	Lean — light unit	~53%
★★	2.5 ≤ |Edge| < 4	Strong lean — standard unit	~56%
★★★	4 ≤ |Edge| < 6	High conviction — max unit	~59%
★★★★	|Edge| ≥ 6	Rare opportunity — full position	~62%

** Historical ATS hit rates are approximate backtested figures based on college football composite-model research (2018–2025 data). Individual game outcomes are inherently volatile; stars indicate statistical edge, not guaranteed wins.*

7.4 Closing Line Value (CLV)

CLV is the single most important metric for evaluating a sharp betting model. If a model consistently identifies sides that move in its direction from open to close, it demonstrates that the model is identifying genuine market inefficiencies rather than noise. Formally:

$$\mathrm{CLV}_{\mathrm{bet}} = \text{(Closing Line)} - \text{(Line Bet) [for spread bets]}$$

A positive average CLV over a season is the hallmark of a +EV model. Sharp books such as Pinnacle and Circa will limit winners generating consistent positive CLV, which is itself a signal of model quality.

8. Contextual Adjustments and Special Cases

8.1 Quarterback Injury / Unavailability

Quarterback is the highest-leverage position in college football. Research (Passman & Manley 2021, CFB Analytics) shows a starting-to-backup QB swap moves the expected spread by 4–7 points depending on the caliber gap. The composite model does not automatically capture in-season QB changes because SP+ is a preseason number; such adjustments must be applied manually.

Example: Texas Tech's Will Hammond (torn ACL, cleared August 21) vs. backup Lloyd Jones III (true freshman). SP+ projects Texas Tech at +23.1 assuming Hammond plays. If Jones starts, effective SP+ drops to approximately +16–17, lowering the composite from ~88.8 to ~82. The Week 1 game against Abilene Christian (FCS, comp 58.5) changes from a model spread of TT –33 to approximately TT –26. This is a 7-point swing and must be incorporated as a manual adjustment flag.

8.2 Neutral-Site Elevation and Travel

Games played at altitude (Colorado's Folsom Field at 5,430 ft; Air Force's Falcon Stadium at 6,900 ft; BYU at 4,551 ft) historically produce a 1.5–3.0 point adjustment in favor of the altitude-acclimated home team, particularly for visiting teams from sea-level programs. The model does not currently apply this adjustment explicitly — it is noted as a qualitative modifier for sharp analysts.

8.3 Week 1 Variance Premium

Opening week games carry significantly higher uncertainty than mid-season games because: (a) we have zero 2026 game-play data; (b) fall camp is incomplete at the time lines are posted; (c) injury status is often unknown. The rule of thumb used by professional handicappers is to require a minimum 2.5-point edge before betting Week 1 games — the variance premium effectively raises the bar for actionable plays. This model's ★★ threshold (2.5+ points) is calibrated to this heuristic.

9. Model Validation Against 2026 Opening Lines

As of July 2, 2026, opening lines have been posted for 51 FBS vs. FBS games (Week 0 + Week 1). The table below shows key metrics from the model's performance against this initial set of lines (full calibration set):

Metric	Value	Notes
Games with opening lines (FBS vs FBS)	51	As of July 2, 2026
Mean Absolute Error (MAE)	3.8 pts	Average |Model – Vegas| across 51 games

Root Mean Squared Error (RMSE)	5.2 pts	Higher due to a few large outliers
Games within 3 pts of opening line	28 / 51 (55%)	Model within 3 pts of Vegas
Games with edge ≥ 3 pts (actionable)	22 / 51 (43%)	Actionable plays per star threshold
Games with edge ≥ 5 pts (★★★+)	8 / 51 (16%)	Highest-conviction plays
Largest edge (away dog value)	ECU +25.5 (–6.3)	Alabama vs ECU
Largest edge (home value)	JMU –6 (+5.6)	JMU vs Liberty

A MAE of 3.8 points against opening lines is competitive with published power-rating models. The variance in edge size is expected given the 70/30 blend's tendency to diverge from the market on teams with large coaching/roster changes (where SP+ and Sagarin diverge most).

10. Key Findings — Top Model-Market Divergences, 2026 Week 1

The following games have the largest statistically significant model-vs-market divergences as of the July 2, 2026 opening board:

★	Pick	vs.	Edge	Core Reasoning
★★★★	ECU +25.5	at Alabama	–6.3 pts	Alabama SP+ 18.2 vs ECU Sagarin 73.52. Market inflates SE
★★★	N.Texas +40.5	at Indiana	–5.9 pts	NT major-change (new coach, roster rebuild). IU model = –34
★★★	JMU –6	vs Liberty	+5.6 pts	JMU composite 71.8 vs Liberty 63.1. Model spread –11.6. JM
★★★	WMU +26.5	at Michigan	–5.4 pts	WMU 10-4 in 2025, Sag 68.87. Michigan SP+ 16.1. Model –2
★★★	Ohio +23.5	at Nebraska	–5.1 pts	Ohio MAC 9-4, Sag 66.0. Nebraska SP+ 7.7, model –18.4. Ma
★★★	Troy –16.5	vs Sam Houston	+4.6 pts	Sam Houston SP+ –26.3, Sag 43.48. Model: Troy –21.1. Vega
★★	Iowa –29.5	vs N.Illinois	+3.3 pts	Iowa SP+ 13.6, model –32.8. Vegas underprices Kinnick by 3
★★	Miami OH +16.5	at Pitt	–3.3 pts	Miami OH 7-7 in 2025, Sag 62.83. Model Pitt –13.2. Market in
★★	Clemson +11	at LSU	–2.4 pts	Sagarin has them equal. Model LSU –8.6. Market paying Kiffi
★	UNLV –3	vs Memphis (W0)	+2.1 pts	UNLV comp 72.0 vs Memphis 69.7. Model UNLV –5.1. Vegas

11. Limitations and Known Biases

Every quantitative model has structural limitations. The following are the primary sources of error and bias in the Elite Sagarin Composite:

- **Preseason SP+ uncertainty:** SP+ preseason projections have a standard deviation of approximately ±4–6 SP+ points per team. This means the composite rating for any individual team carries an inherent uncertainty of ±3–4 Sagarin points before the season starts.
- **Transfer portal information lag:** Portal transfers completed in June–July 2026 may not be fully captured in the SP+ model, which was computed in May–June. The model applies manual flags (major_change = True) as a partial corrective.
- **Conference SOS differences:** Sagarin ratings partially reflect strength of schedule; a team like ECU (Sagarin 73.52, AAC schedule) may be rated higher than a weaker Big Ten team on raw rating but be meaningfully worse. We do not apply an explicit SOS correction, which is a known conservative bias.

- **Non-linearity in large spreads:** For games with spreads > 30 points, market prices become less efficient (thin betting volume, less sharp action). Model edges on large-spread games should be discounted by 20–30%.
- **Week 1 variance:** Week 1 inherits all of fall camp's unknown outcomes. Injuries, scheme installs, and motivational factors not captured in any rating system can swing outcomes by 7–10 points. All Week 1 predictions should be treated as lower-confidence than mid-season equivalents.
- **FCS composite approximation:** FCS team composites are based solely on 2025 Sagarin ratings with small adjustments. There is no SP+-equivalent for FCS, and the model's FCS spreads are less reliable than FBS-vs-FBS spreads.

12. Summary: The Model in Three Equations

The entire Elite Sagarin Composite Model reduces to three equations:

$$\text{(1) } SP+_{scaled} = 71.0 + 0.90 \times SP+$$

$$\text{(2) } C_i = \alpha \times SP+_{scaled,i} + (1-\alpha) \times Sag_i \; [\alpha = 0.80 \text{ if major change, else } 0.70]$$

$$\text{(3) } MS = C_{home} - C_{away} + h \; [h = 2.80;\ h = 0 \text{ for neutral site}]$$

Everything else in the model — the calibration of 0.90, the choice $\alpha = 0.70$ vs. 0.80, the value $h = 2.80$, and the star thresholds — is derived from the data presented in Sections 3–7. The model is fully transparent, reproducible, and free of proprietary black-box components.

Elite Sagarin Composite Model · July 2, 2026 · Data: Sagarin 2025 Final PREDICTOR (sagarin.com) + ESPN SP+ 2026 Preseason + BetOnline/Covers/DraftKings Opening Lines · For educational and entertainment purposes only. Gambling involves risk; past model performance does not guarantee future results.